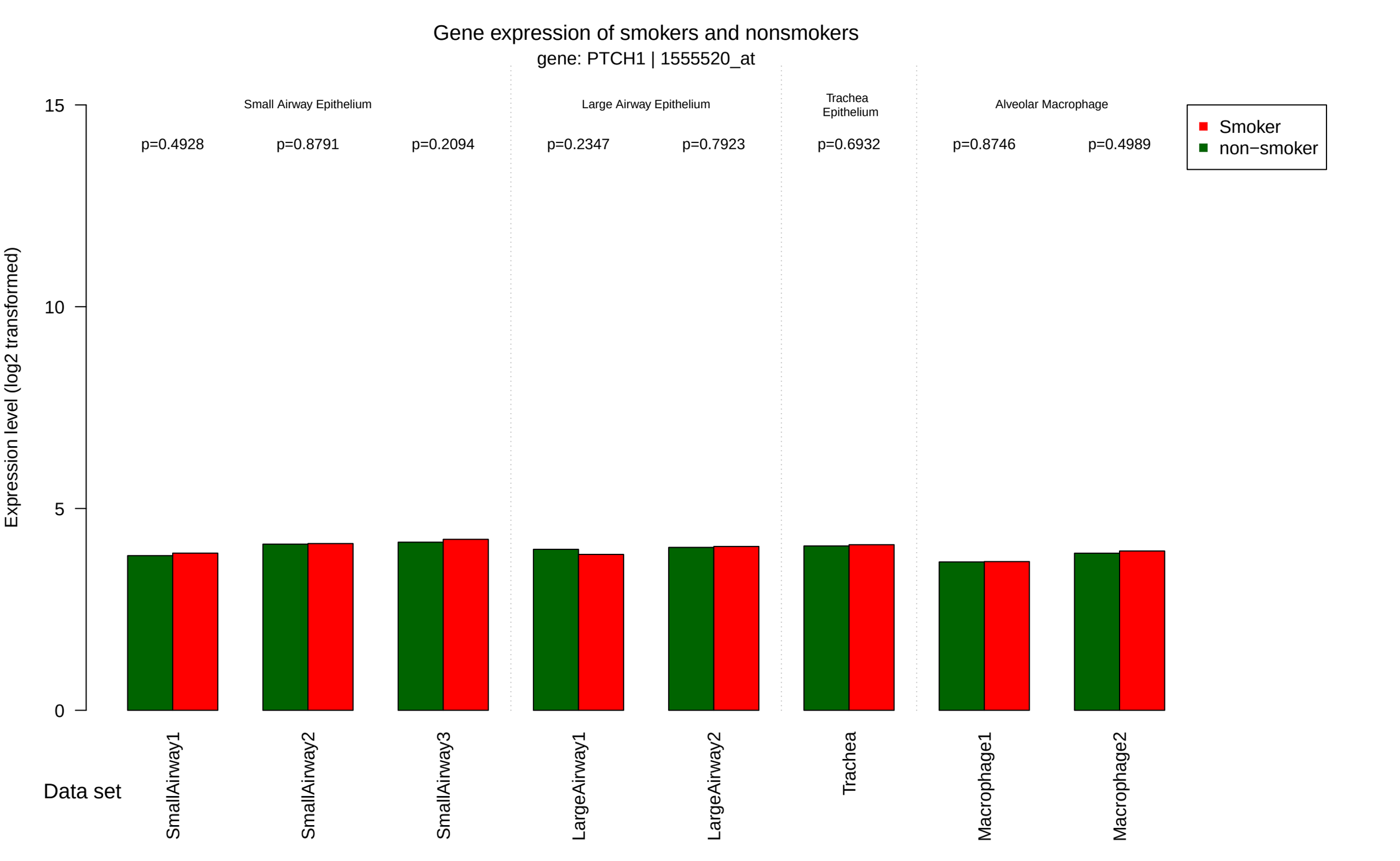
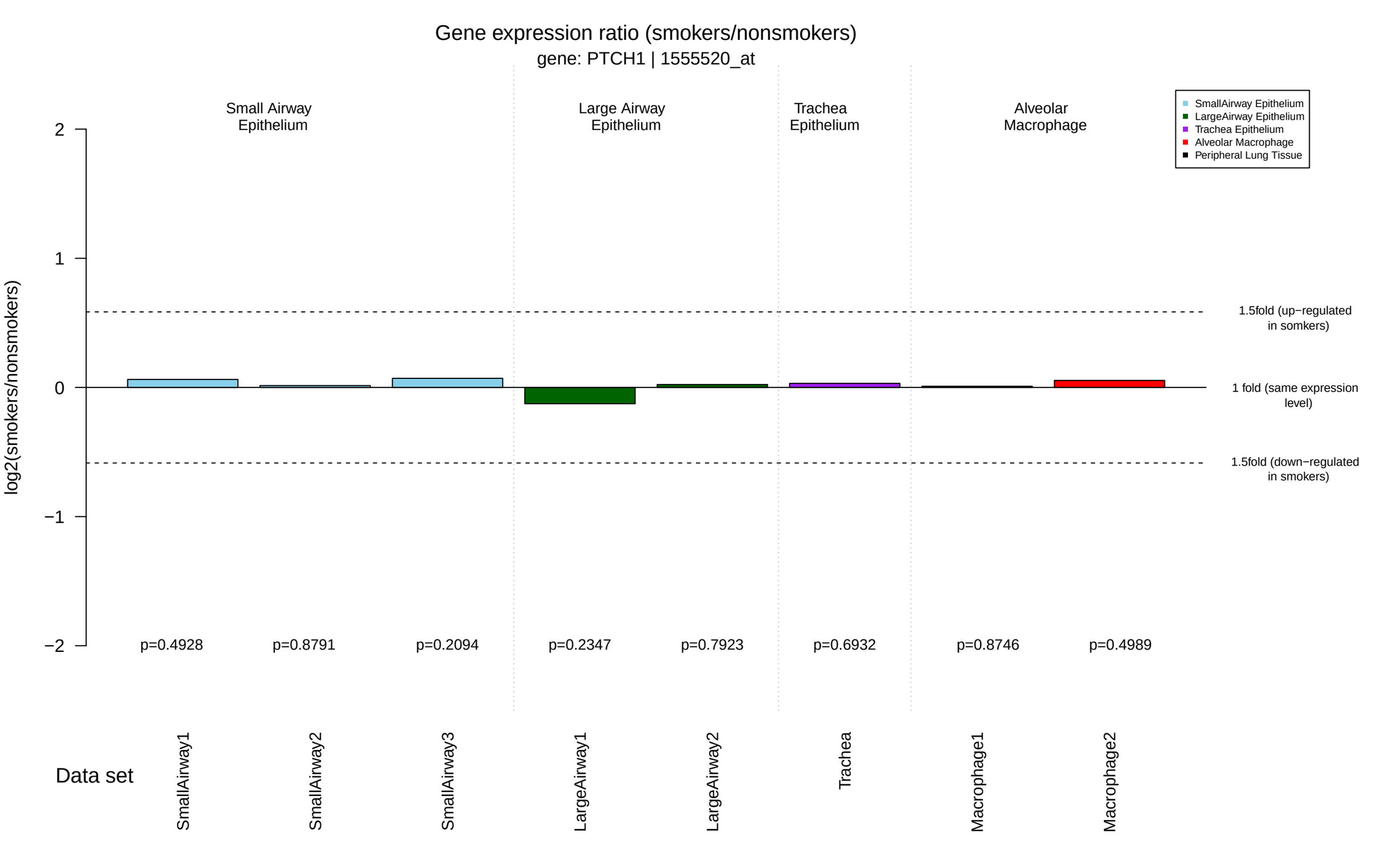
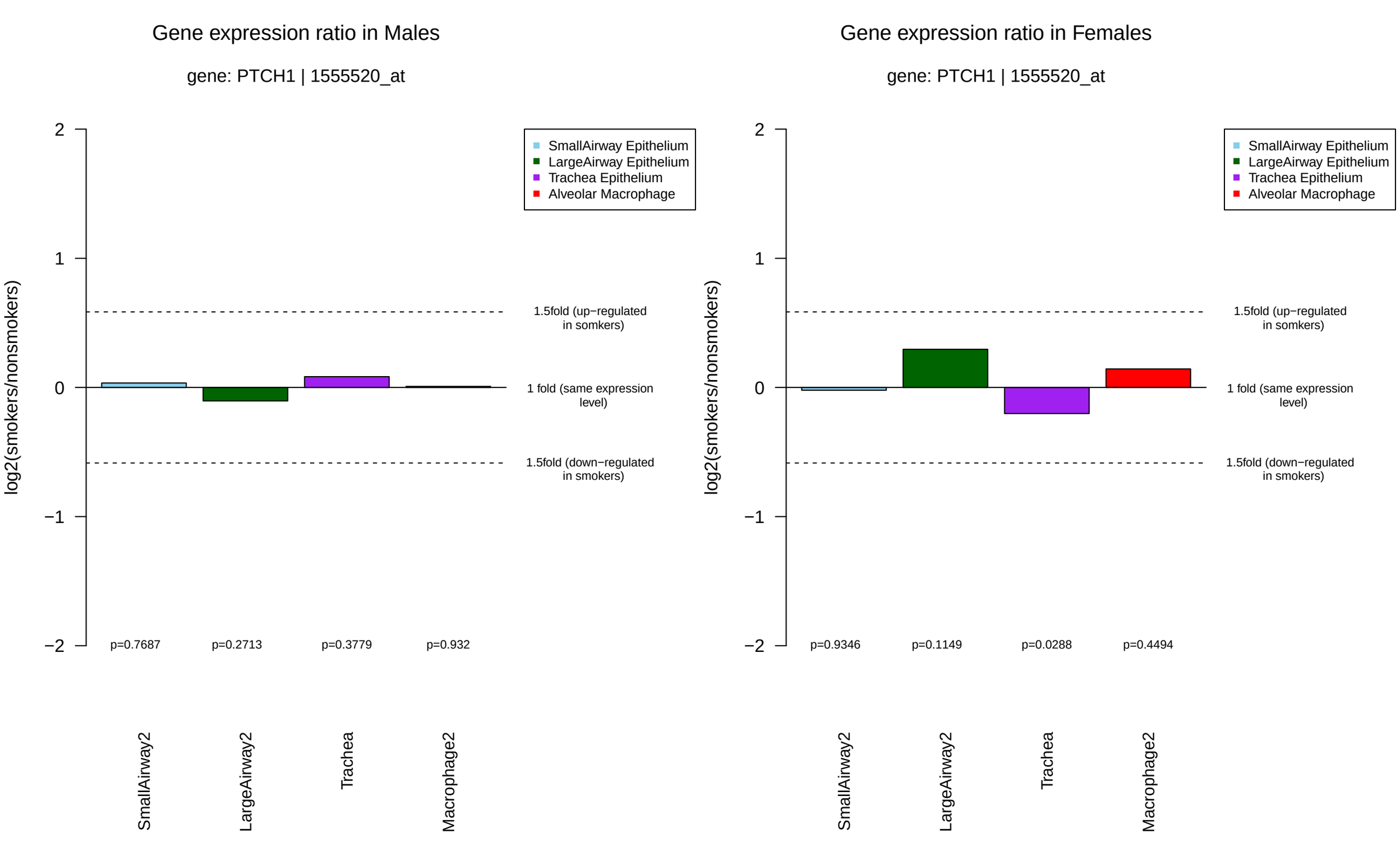
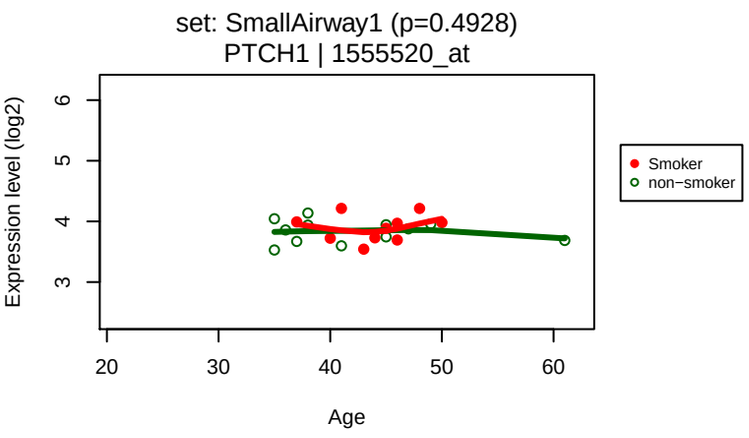
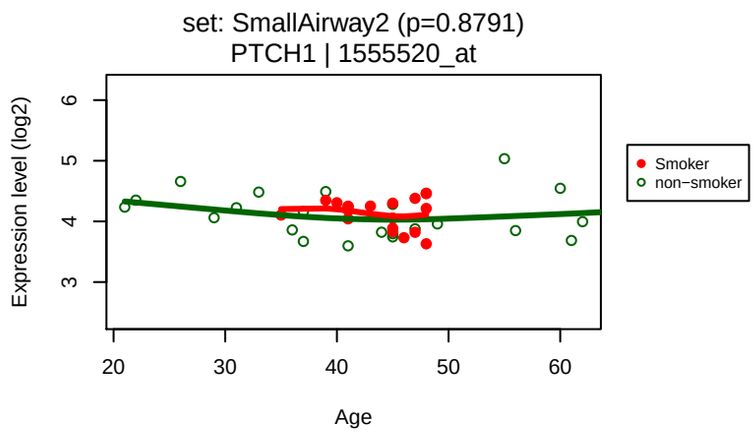
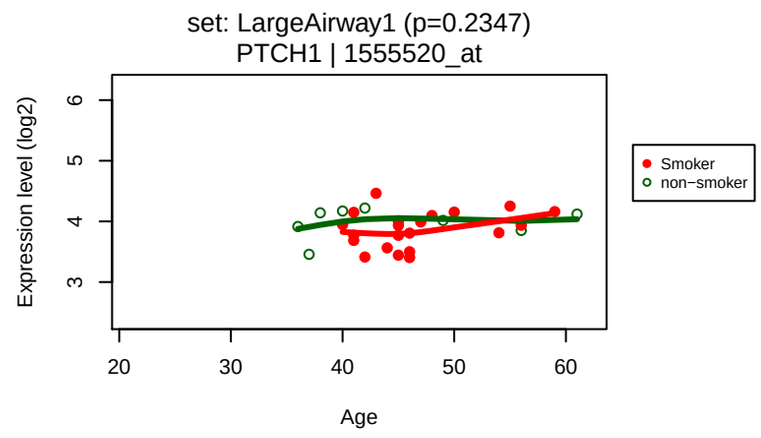

Gene expression of smokers and nonsmokers
gene: PTCH1 | 1555520_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4928
p=0.8791
p=0.2094
p=0.2347
p=0.7923
p=0.6932
p=0.8746
p=0.4989
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

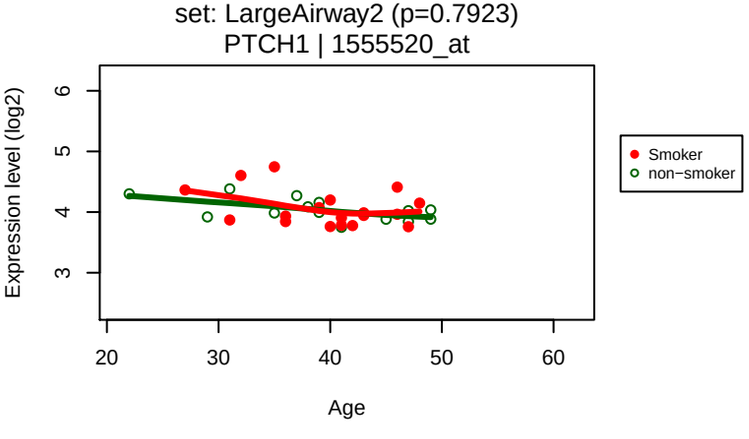

Gene expression ratio (smokers/nonsmokers)
gene: PTCH1 | 1555520_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4928
p=0.8791
p=0.2094
p=0.2347
p=0.7923
p=0.6932
p=0.8746
p=0.4989
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

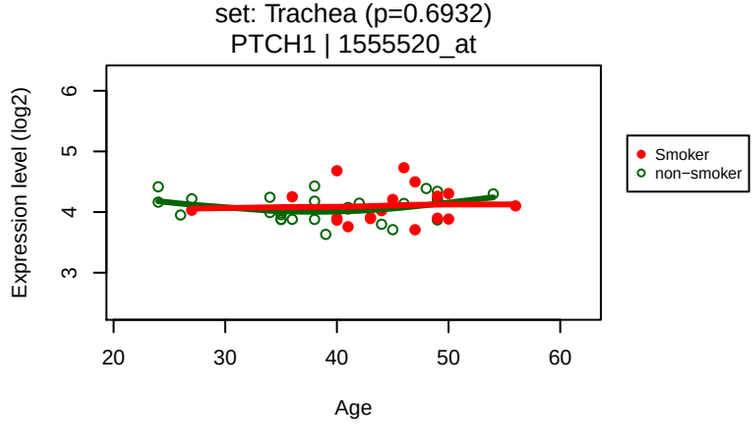

Gene expression ratio in Males
gene: PTCH1 | 1555520_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7687
p=0.2713
p=0.3779
p=0.932
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTCH1 | 1555520_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9346
p=0.1149
p=0.0288
p=0.4494
SmallAirway2
LargeAirway2
Trachea
Macrophage2

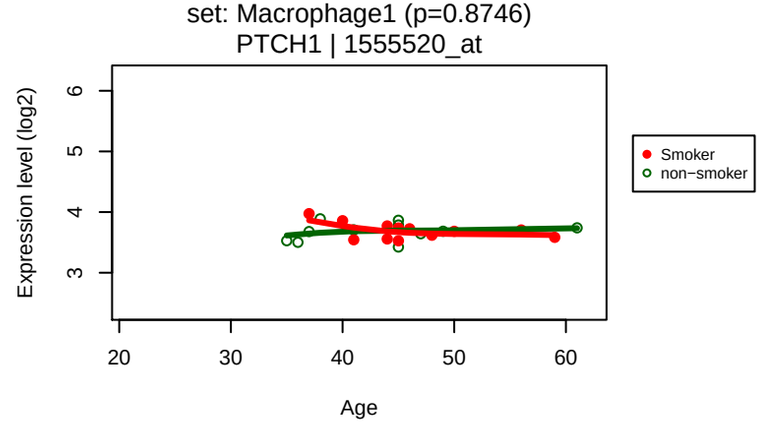

set: SmallAirway1 (p=0.4928)
PTCH1 | 1555520_at
Expression level (log2)
Age
Smoker
non−smoker

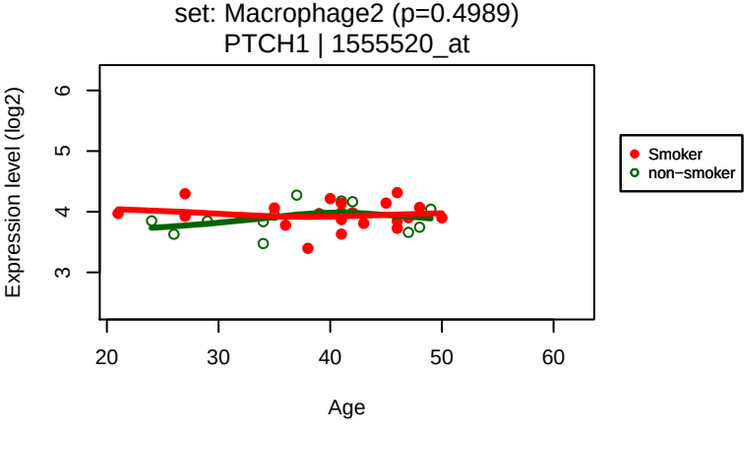

set: SmallAirway2 (p=0.8791)
PTCH1 | 1555520_at
Expression level (log2)
Age
Smoker
non−smoker

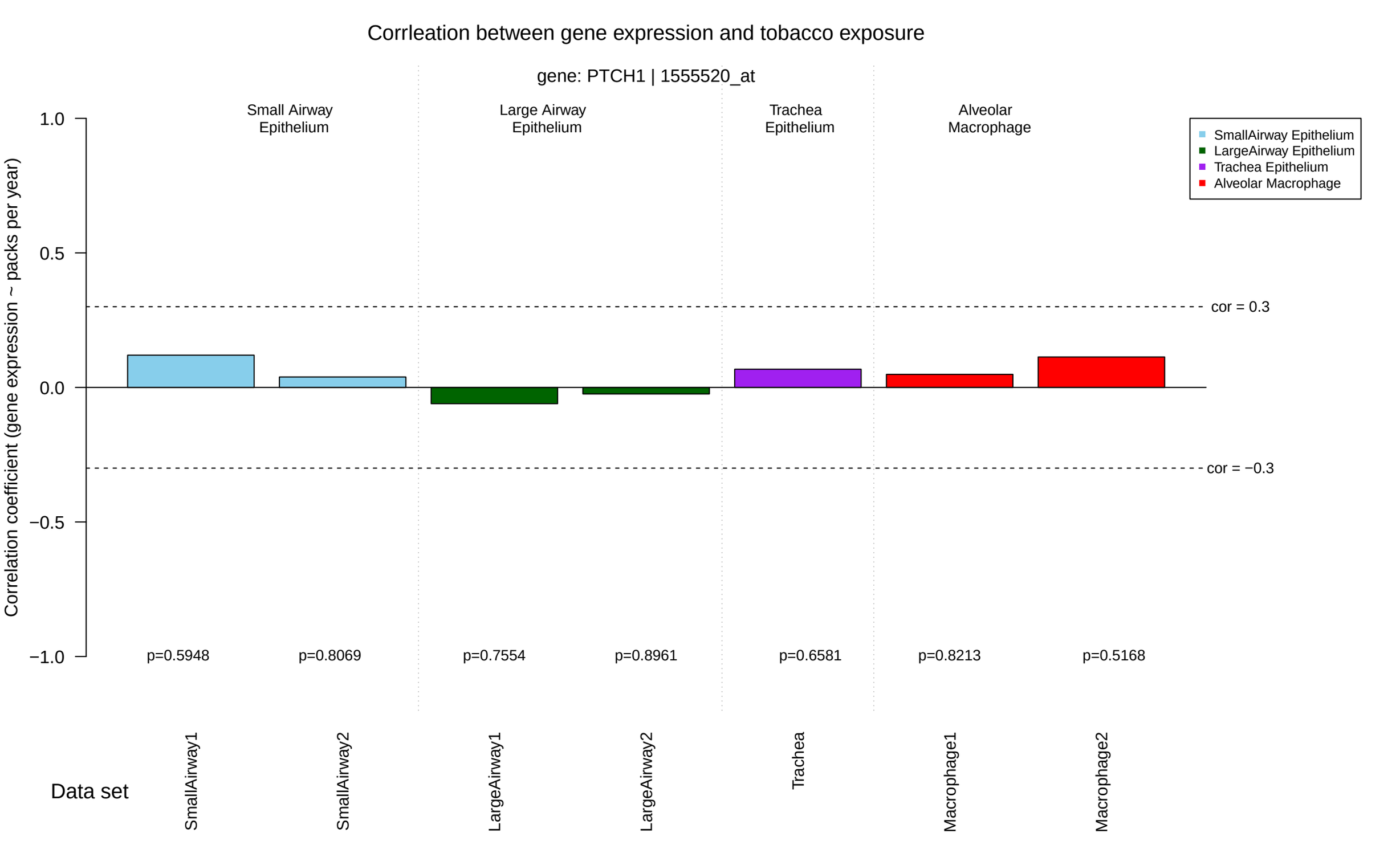

set: LargeAirway1 (p=0.2347)
PTCH1 | 1555520_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7923)
PTCH1 | 1555520_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6932)
PTCH1 | 1555520_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8746)
PTCH1 | 1555520_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4989)
PTCH1 | 1555520_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTCH1 | 1555520_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5948
p=0.8069
p=0.7554
p=0.8961
p=0.6581
p=0.8213
p=0.5168
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2